

Composite coating of xanthan gum with sodium nitroprusside alleviates the quality deterioration in strawberry fruit

Anuja Gautam^{a,*}, P.P.S. Gill^a, NavPrem Singh^a, S.K. Jawandha^a, Rachna Arora^a, Arashdeep Singh^b, Ajay^a

^a Department of Fruit Science, Punjab Agricultural University, Ludhiana, 141004, Punjab, India

^b Department of Food Technology, Punjab Agricultural University, Ludhiana, 141004, Punjab, India

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ABSTRACT

Strawberry is an important and highly nutritious fruit crop that undergoes various postharvest qualitative losses. Therefore, the present study aimed to assess the synergistic effect of xanthan gum (XG) coatings with sodium nitroprusside (SNP) on the antioxidant defense system of strawberries during 16 days of storage by dipping method. The results on the 16th day of storage revealed that in comparison to control, the composite coating of XG 0.50 % + SNP reduced fruit weight loss (31.35 %) and spoilage incidence (87.22 %), along with effectively delayed respiration rates (16.44 %). Furthermore, the coatings maintained the biochemical properties of fruit including soluble solids content (7.05 %), titratable acidity (30.05 %), total flavonoid content (31.05 %), total anthocyanins content (17.59 %) and total phenols content (38.20 %) as compared to untreated fruit. The XG 0.50 % + SNP coated fruits also registered significantly ($p < 0.05$) higher total antioxidant activity (TAA) and antioxidant enzyme activities including catalase, peroxidase, and phenylalanine ammonia-lyase throughout the storage. Additionally, functionalized coating of XG with SNP suppressed the activity of polyphenol oxidase (PPO). This study proved that the application of coating of XG 0.50 % + SNP 2.0 mM is a potential treatment to preserve the postharvest quality of strawberries.

1. Introduction

Strawberry (*Fragaria × ananassa* Duch) is the most common berry consumed, both for fresh and processing purposes. It is a rich source of various plant metabolites including minerals, flavonoids, and polyphenol compounds, many of which are natural antioxidants and contribute to its high nutritional value (Bahmani, Razavi, Mortazavi, Gohari, & Juárez-Maldonado, 2022). The presence of these compounds synergistically promotes human health and prevents diseases. Strawberry is a non-climacteric fruit, and for obtaining maximum market quality requires the harvesting of fruit at full maturity. Due to high respiration rate, moisture loss, delicate texture and sensitivity to mechanical damage, strawberries are highly perishable and have a short shelf-life (Tahir et al., 2018). Furthermore, reactive oxygen species (ROS) cause oxidative stress in the fruit, which in turn reduces its postharvest quality and consumer acceptability because of the oxidation of plant metabolites (Xu et al., 2022). Strawberry fruits are also subjected to the attack of various mycotoxin-producing fungi which not

only leads to visible spoilage but also contaminates the strawberry juices and jams (Fernández Cruz, Mansilla, & Tadeo, 2010). Therefore, the strawberry industry faces a major challenge in postharvest handling, so finding simple ways to maintain fruit freshness is becoming crucial. To date, several postharvest methods to enhance the storage life of fresh strawberries have been investigated, including refrigeration, modified atmosphere packaging (MAP), controlled atmospheres, radiation and synthetic fungicides (Iniguez-Moreno et al., 2023). Although refrigerated storage is the most commonly employed method however, it may lead to heavy postharvest decay due to chilling injury, internal browning, pulp discoloration and tissue corrosion (Wang, Ding, Zhou, Gao, & Fan, 2016). Moreover, some of these methods are not commercially feasible due to the high cost of the technology and low customer preference. The common method of controlling postharvest decay is the use of synthetic fungicides, but these may leave residues that are harmful to humans and the environment. The strawberry industry demands a green-labelled and cost-effective approach that can effectively minimize postharvest losses and maintain maximum fruit quality. As an

* Corresponding author.

E-mail address: anuja-2297001@pau.edu (A. Gautam).

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alternative to synthetic preservative materials and expensive post-harvest technologies, edible coatings have proven to be an effective option since they create a protective layer that acts as a barrier between the fruit surface and the outside atmosphere, thereby reducing the respiration rate and oxidative stress of the fruit (Adhikary, Gill, & Jawandha, 2023; Ajay et al., 2024). Coatings, composed of natural elements like lipids, carbohydrates, proteins and other food-grade substances, fulfil roles such as retaining moisture, preserving freshness, extending shelf life, antioxidant and antimicrobial properties (Kumar, Prasad, et al., 2023). The uncoated strawberry fruits incurred 67.37 % higher weight loss as compared to the strawberries treated with a composite coating of aloe vera gel and carnauba wax, stored at 4 °C (Álvarez-Barreto et al. (2023)). Moreover, the use of MAP instead of macro-perforated packaging in strawberries leads to 17 % reduction in average losses (Matar et al., 2020). As an edible coating, xanthan gum is readily available and a high molecular weight (2×10^6 – 5×10^7 Da) polysaccharide obtained from the aerobic fermentation of the bacterium *Xanthomonas campestris*. Because of its high potency, it is utilized as a commercial hydrocolloid acting as a thickening agent and stabilizer in the food industry (Zheng et al., 2019). Xanthan gum is a branched polysaccharide whose backbone is composed of cellobiose as repeating units and side-chain composed of a D-mannose, D-glucuronic acid and D-mannose trisaccharide complex which are attached to alternate glucose units by α -1, 3 linkages. Terminal D-mannose at its C-4 is attached to the pyruvic acid residue, linked via the keto group at C-4 and C-6 positions. The D-mannose unit linked to the glucose main chain has its C-6 position replaced with an acetyl group (Petri, 2015). Xanthan gum's unique rheological properties allow it to create highly viscous solutions in a wide range of pH and temperature (Zhang, Jiang, Ling, Ouyang, & Wang, 2021). Furthermore, it has many advantages over film-forming properties, including its safety for humans, efficiency, visibility, colorlessness, odorlessness, cheapness, availability, and high level of biodegradability (Elella et al., 2021). Previous studies indicated that XG-based edible coatings preserved the antioxidant activity, delayed weight loss and retained the firmness and quality of Elamoura guava (Gad & Zagzog, 2017), mango (Prasad et al., 2022) and fresh-cut pears (Sharma & Rao, 2015). A combination of an edible coating and functional substances (antimicrobial, antioxidants and nutraceuticals) to boost food preservation is the emerging strategy (Mitelut et al., 2021). Sodium nitroprusside (SNP), a well-known synthetic compound is the donor of nitric oxide (NO), a highly reactive free radical gas that acts as an antioxidant and signaling molecule in activating ROS scavenging enzymes (Sahu, Barman, & Singh, 2020; Siddiqui, Al-Whaibi, & Basalah, 2011). The enzymes ACC synthase and ACC oxidase bind NO, which slows down the rate of respiration and ethylene production in fruits (Rümer, Gupta, & Kaiser, 2009). It is possible that NO might contribute to the postharvest storage protocol since it retards the senescence process (Adhikary, Gill, Jawandha, Bhardwaj, & Anurag, 2021). NO being a biological messenger in plants contributes indirect health benefits to humans by improving the flavours, anthocyanin content, soluble sugars, mineral content, amino acids and vitamin-C in Blood Orange fruits (Zhang, Liu, et al., 2022). Similarly, several researchers stated that the use of sodium nitroprusside was effective in minimizing postharvest losses in fruits like apple (Chen et al., 2019), lemon (Kaur, Jawandha, Gill, Grewal, & Singh, 2022), blueberry (Ge et al., 2019) and pear (Adhikary et al., 2021). Therefore, the incorporation of SNP into XG could be a potent therapy for minimizing postharvest losses in strawberries. The gas barrier properties of XG along with the antioxidant capacity of SNP can provide new insights for the prevention of post-harvest strawberry losses and maintain its nutritional qualities while considering its eco-friendly, health-friendly and sustainable nature. However, to the best of our knowledge, no study is available on the combined effect of XG and SNP to maintain the postharvest quality of strawberries through the regulation of non-enzymatic and enzymatic antioxidants of stored strawberries. In this context, investigations were made to quantify the effect of different formulations of XG coatings,

alone or enriched with SNP on the antioxidant defense system and various quality attributes of strawberries under low-temperature storage.

2. Material and methods

2.1. Fruit material

Mature 'Winter Dawn' strawberry fruits free from bruises and blemishes were harvested during the early morning hours from the Experimental Farm of Punjab Agricultural University, Ludhiana (India) in February 2023. The fruits were harvested with the help of fruit clip-pers and retained small peduncles and calyxes. Thereafter, the fruits were immediately shifted to the Post Harvest Laboratory and were then sorted based on shape, size and any visible injuries, followed by disinfection with 1.0 % sodium hypochlorite and rinsed with water (Tahir et al., 2018). After air-drying, the fruits were divided into six equal lots, keeping 225 fruit per lot to get three replications.

2.2. Formulation of edible coatings and treatments

For the preparation of XG coatings, 2.5 and 5.0 g of xanthan powder (Xanthan gum pure (food grade), product code: 26,883, Central Drug House (P) Ltd. India) was dissolved in distilled water with continuous stirring on a magnetic stirrer (Spinot Digital 6040, Tarsons, India) at 750 rpm for 1 h to obtain XG 0.25 % and XG 0.50 % formulation. For the blended coating formulations of XG and SNP, 2.0 mM SNP was added to XG coatings with continuous stirring on a magnetic stirrer to obtain XG 0.25 % + SNP 2.0 mM and XG 5.0 % + SNP 2.0 mM concentrations of coatings. For sole SNP coating, 2.0 mM SNP was prepared with distilled water. In all treatments, 1.0 % glycerol (Sisco Research Laboratories Pvt. Ltd., 99 % purity) was used as a plasticizer. Afterwards, the fruits were immersed into different coating formulations, XG 0.25 and 0.50 % in combination with SNP 2.0 mM and sole treatment of XG 0.25 %, XG 0.50 % and SNP 2.0 mM. The uncoated fruits treated with distilled water were considered as control. The fruits were then dried and packed in plastic punnet boxes stored at 4 ± 1 °C at 90–95 % RH and analyzed for various quality and enzymatic attributes (Fig. 1) at 0, 4, 8, 12, and 16 days (d) of storage.

2.3. Fourier transform infrared (FT-IR) analysis

FTIR graphical spectra of the edible coatings were obtained on FT-IR spectrometer (Cary 630 FTIR, Agilent Technologies, USA) with 20 scans to analyse the incorporation of SNP into XG coatings. The samples were directly poured on the ATR (attenuated total reflectance) crystal after clearing the background with methanol. The wavenumber range in which the spectra were recorded was 500 – 4500 cm^{-1} at 20 °C (Sharma et al., 2023).

2.4. Weight loss

Using a digital weighing balance (EK-6100i, A&D, Japan), the fruits were weighed at different storage intervals. Loss in weight was expressed in terms of percentage and determined using equation (1).

$$\text{Loss in weight (\%)} = \frac{\text{Initial fruit weight} - \text{Final fruit weight}}{\text{Initial fruit weight}} \times 100 \quad (1)$$

2.5. Respiration rate

The respiration rate was measured using the closed-system method described by de Oliveira Filho et al. (2022). To measure respiration rates, approximately 250 g of strawberries were kept in hermetically sealed glass jars of 1000 ml capacity. The CO_2 concentration was measured from an impermeable glass jar using a headspace gas analyser

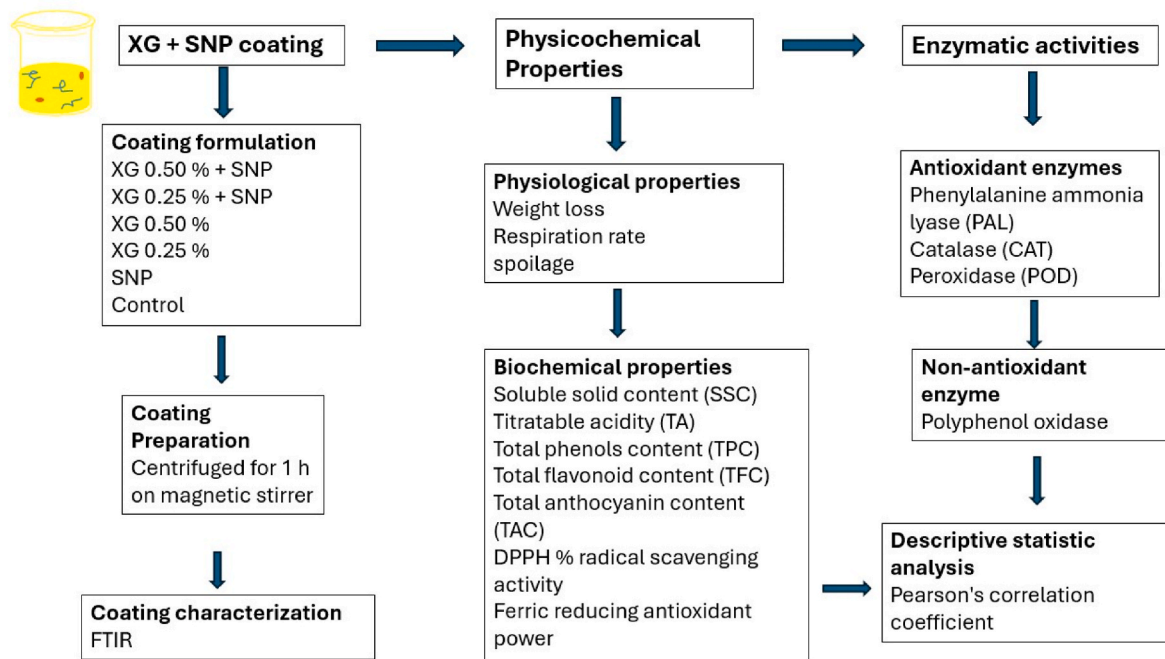


Fig. 1. A schematic flow diagram showing the analysis performed during the study.

(Gaspac Advance Systech, GS3/P, Systech Instruments, UK) for 5 h at 4 °C. The respiration rate was expressed as $\text{ml CO}_2 \text{ kg}^{-1} \text{ FW h}^{-1}$.

2.6. Spoilage

Fruit spoilage was calculated based on the number of fruits spoiled at each sampling interval (Tahir et al., 2018). Calculations of fruit spoilage percentage were performed using equation (2).

$$\text{Fruit spoilage (\%)} = \frac{\text{Number of spoiled fruits}}{\text{Total number of fruits}} \times 100 \quad (2)$$

2.7. Soluble solids content (SSC) and titratable acidity (TA)

A digital refractometer Pal-1 by ATAGO Co., L., Tokyo, Japan) was used to measure the fruit soluble solids content, expressed in °Brix at 20 °C. Before observations were made, the refractometer was calibrated with distilled water to read zero. The TA of strawberry fruit was determined by following the titration method (Saleem et al., 2021). The 5.0 mL of fruit juice was homogenized with 20 mL of distilled water and titrated against 0.1 N NaOH, while phenolphthalein was used as an indicator and the results were expressed in percent.

2.8. Total phenols content (TPC)

A procedure similar to the one followed by Sinha, Gill, Jawandha, Kaur, and Grewal (2021) was used to determine the fruit TPC. The TPC was quantified using the Folin-Ciocalteu reagent protocol. Briefly, 1.0 g of fruit tissue was blended with 3.0 mL of 80 % methanol, followed by centrifugation for 20–25 min at $10,000 \times g$ at 4 °C and the 500 μL of supernatant was collected and diluted to 10 mL with distilled water, from which 100 μL of the extract was used to estimate the TPC. Further, 100 μL extract was introduced in the test tube along with 1.5 mL of freshly made Folin-Ciocalteu reagent and 4.0 mL of 20 % Na_2CO_3 . The final volume was made 10 mL with distilled water, and the reaction mixture was dark incubated for 30 min. The absorbance was recorded at 738 nm (Genesys 50, UV-Visible Spectrophotometer, Thermo Scientific) and the outcome was expressed as g gallic acid equivalent per kg fresh weight.

2.9. Total flavonoid content (TFC)

The TFC was estimated using the aluminium chloride method described by Sharma, Bhatia, and Kaur (2018). In 80 % (v/v) methanol, homogenized fruit pulp was centrifuged for 15 min at $10,000 \times g$. Further, the 0.5 mL supernatant of methanol extract was evaporated to dryness, followed by the addition of 1.0 mL of distilled water and 2.5 mL of AlCl_3 (0.1 M in 80 % methanol). At room temperature, the reaction mixture was kept for 10–15 min, and the yellow color produced was measured at 420 nm and expressed as g rutin equivalents (RE) kg^{-1} fresh weight.

2.10. Total anthocyanins content (TAC)

The TAC was extracted and estimated by the method of Gol, Patel, and Rao (2013). Briefly, the fruit pulp was combined with 20 mL of extracting solvent (95% ethanol: 1.5 N HCl, 85:15) and left overnight at 4 °C. Thereafter, samples were filtered into a flask and covered with aluminium foil. An extractor solution was utilized to remove the remaining pigments from the residue. The filtrate was made into a final volume of 100 mL with the same solvent, and absorbance was recorded at 535 nm using a digital spectrophotometer and the results were expressed as mg kg^{-1} .

2.11. Total antioxidant activity (TAA)

The antioxidant activity of strawberries at each storage interval was detected by the free radical scavenging capacity DPPH and ferric reducing antioxidant power (FRAP). DPPH was determined using 2, 2-diphenyl-1-picryl-hydrazyl radical-scavenging methodology followed by Sinha et al. (2021). DPPH was determined by mixing 0.1 mL of methanolic extract of fruit with 0.29 mL of a methanolic solution of DPPH (0.1 mM). After 30 min of incubation at 25 °C of the resulting reagent, the absorbance of the reagent was determined at 517 nm and the results were expressed as a percentage of radical scavenging activity. The FRAP assay was performed using the methodology followed by Zhang, Li, et al. (2022). For analysis, 20 μL of sample extract was added to 1.8 mL of the freshly prepared FRAP reagent (300 mM acetate buffer

with pH 3.6, 10 mM 2,4,6-tri (2-pyridyl)-s-triazine and 20 mM $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ solution in the ratio 10:1:1). The freshly prepared mixture was then incubated at 37 °C for 30 min in a water bath. Further absorbance values were measured at 593 nm. The outcome was expressed in mmol kg^{-1} fresh weight. A mixture of all reagents except the sample was used as a control for DPPH and FRAP assay.

2.12. Antioxidant enzymatic activities

2.12.1. Phenylalanine ammonia lyase (PAL)

The method of Kumar, Hunjan, Kaur, Dhillon, and Singh (2017) was slightly modified to quantify the PAL activity. Briefly, 0.1 g of fruit pulp was homogenized in 2.0 mL of tris-HCl buffer (0.1 M with pH 7.5) in a pre-chilled pestle and mortar. Thereafter, the homogenized samples were centrifuged at $10,000 \times g$ for 30 min at 4 °C following which the supernatant was collected and used for further estimation. To assay the estimation of PAL activity, 2.5 mL of L-phenylalanine (0.03M) prepared in sodium borate buffer with pH 8.8 and 100 μL of enzyme extract supernatant were added in a test tube, followed by incubation at 37 °C for 60 min. Thereafter, the reaction was halted by the addition of 300 μL of HCl (5 N) and the absorbance at 290 nm was recorded. As a control, another set of reactions was terminated immediately without incubation. The outcome of the PAL enzyme was expressed in U mg^{-1} protein.

2.12.2. Catalase activity (CAT) and peroxidase activity (POD)

A 0.1 g sample of fruit tissue was homogenized in a pre-chilled pestle and mortar with 2.0 mL of potassium phosphate buffer with a pH of 7.5 containing 1.0 % of polyvinylpyrrolidone (PVP), β -mercaptoethanol (10 mM) and EDTA (10 mM) further centrifuged for 30 min at $10,000 \times g$ at 4 °C for enzyme extraction. To estimate CAT activity, 200 μL of enzyme extract and 1.8 mL of 0.05 M sodium phosphate buffer (pH 7.5) were added to the cuvette. A solution of 1.0 mL H_2O_2 solution was added to start the reaction. This solution was made by diluting 0.2 mL of H_2O_2 (30%) into 25 mL of 50 mM sodium phosphate buffer (pH 7.5). One unit of CAT activity was defined as a decrease of 0.01 in absorbance per min at 240 nm and expressed as Units (U) mg^{-1} protein. The POD activity was measured at 470 nm by estimating the oxidation of guaiacol in the presence of H_2O_2 . The estimation mixture contained 3.0 mL of guaiacol (0.05 M), 200 μL of enzyme extract, and 0.1 mL of H_2O_2 (0.8 M). To initiate the reaction, H_2O_2 was added and the rate of change in absorbance was measured. The outcome was expressed as U mg^{-1} protein (Kaur, Grewal, Singh, & Kaur, 2017).

2.13. Polyphenol oxidase (PPO)

PPO enzyme activity was estimated as per the method followed by Sinha et al. (2021). For the extraction of the PPO enzyme, 0.2 g of fruit tissue was completely homogenized in 2.0 mL of 0.1 mM phosphate buffer (pH 6.8) in a pre-chilled pestle and mortar and further centrifuged for 30 min at $10,000 \times g$ at 4 °C. After centrifugation, the supernatants obtained from extraction were used for PPO enzyme estimation. For estimation, 500 μL enzyme extracts were added to 1 mL of 0.1 mM phosphate buffer (pH 6.8) and 0.5 mL of 0.1 mol l^{-1} catechol. One unit of PPO activity was defined as the increase of 0.01 in absorbance per min at 410 nm and expressed as U mg^{-1} protein.

2.14. Data analysis

The experiment was laid out in a completely randomized design (factorial: treatment and storage interval) with three replicates. Data were analyzed by two-way analysis of variance using the statistical software SPSS (version 22.0, $p < 0.05$). The interaction between treatments as factor 1 and storage intervals as factor 2 was examined and the means were differentiated with the least significant difference (LSD) test. Differences among treatments were considered statistically significant at $p < 0.05$ level of significance. Pearson's linear correlation

coefficient (r) among the various antioxidant parameters was carried out using the statistical software R studio Version March 4, 5958.

3. Results

3.1. Fourier transform infrared (FT-IR)

Using FT-IR method the authenticity of the coatings and possible interactions between polymer chains was determined (Fig. 2). The characteristic spectra of xanthan gum coating were found at 3744, 3332, 2305, 2115, 1791, 1636 and 1413 cm^{-1} . The SNP sole coating showed a characteristic peak at 3647, 3332, 1943 and 1636 cm^{-1} . Additionally, new spectral bands in combined xanthan gum and SNP coatings were obtained at 1941 cm^{-1} along with the disappearance of some vibrations from 1871 to 1722 cm^{-1} . Furthermore, the composite coatings showed vibrations from 3567 to 3690 cm^{-1} , 1645 to 1941 and 1558 to 1339 cm^{-1} .

3.2. Weight loss (%)

During the entire storage period of 16 days, a continuous increase in fruit weight loss was observed in all the treatments (Table 1). Throughout the storage period, coated fruit recorded lower weight loss as compared to control. SNP combined XG (0.50%) coated fruits showed a lesser increase in weight loss than its sole XG and sole SNP coated fruits. However, on the 4th day, XG 0.25 % + SNP showed non-significant higher weight loss than XG 0.25 % coated fruits and afterwards, XG 0.25 % + SNP showed lower weight loss than XG 0.25 % coated fruits. On 8th, 12th and 16th day of the storage, XG 0.50 % + SNP coated fruits exhibited 7.86, 10.08 and 7.00 %, respectively lower weight loss than XG 0.50 % coated fruits. The maximum loss in weight at the last sampling was observed in control fruits followed by SNP treated fruits with a value of 6.05 % and 5.72 % respectively. On the 16th day, the weight loss in XG 0.50 % + SNP strawberries was significantly ($p < 0.05$) lower by 31.35 % than the control strawberries.

3.3. Respiration rate ($\text{ml CO}_2 \text{ kg}^{-1} \text{ h}^{-1}$)

Strawberry fruits stored at 4 °C showed an increasing trend in respiration rate throughout the storage period (Table 1). During the entire period of storage, the sole XG 0.25 % coated strawberries exhibited a lower respiration rate in comparison to XG + SNP coated fruits. On the 8th and 12th day, XG 0.50 % + SNP coated fruits exhibited 39.06 and 12.25 %, respectively lower respiration rates than alone SNP

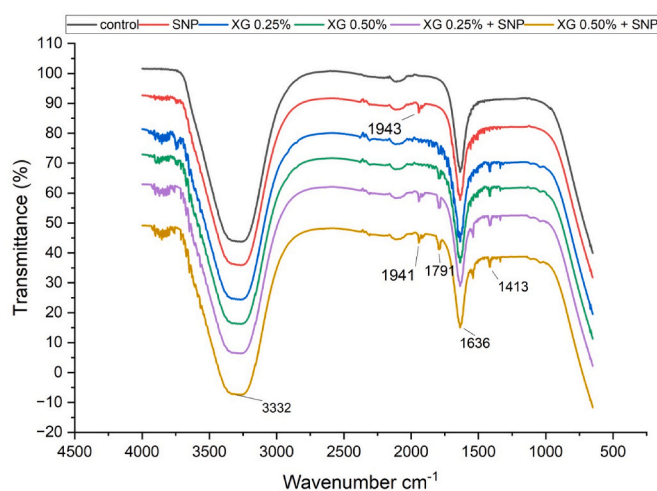


Fig. 2. FTIR spectra of different coatings applied on strawberry fruits. XG in the graph legend denotes xanthan gum while SNP denotes sodium nitroprusside.

Table 1

Effect of different concentrations of XG and SNP (2.0 mM) on weight loss, respiration rate and spoilage on strawberry fruits during storage. Data presented are the mean values $\pm$ SE of 3 replications ($p < 0.05$). Different letters indicate significant differences at ($p < 0.05$) using least significant difference.

Parameter	Treatment	Storage time (d)				
		0	4	8	12	16
Weight loss (%)	Control	0.0 $\pm$ 0 ^q	1.34 $\pm$ 0.16 ^{m-p}	2.98 $\pm$ 0.18 ^{s-j}	4.85 $\pm$ 0.18 ^e	6.05 $\pm$ 0.17 ^a
	XG 0.50 % + SNP		0.75 $\pm$ 0.18 ^{p-q}	1.71 $\pm$ 0.16 ^{l-o}	3.20 $\pm$ 0.15 ^{s-i}	4.41 $\pm$ 0.19 ^e
	XG 0.25 % + SNP		1.05 $\pm$ 0.13 ^{n-p}	2.19 $\pm$ 0.18 ^{i-m}	3.68 $\pm$ 0.16 ^{fg}	5.13 $\pm$ 0.16 ^{cd}
	XG 0.50 %		0.82 $\pm$ 0.14 ^{o-q}	1.85 $\pm$ 0.19 ^{k-n}	3.54 $\pm$ 0.16 ^{f-h}	4.73 $\pm$ 0.18 ^{de}
	XG 0.25 %		0.90 $\pm$ 0.15 ^{i-l}	2.40 $\pm$ 0.19 ^{op}	4.19 $\pm$ 0.19 ^{ef}	5.36 $\pm$ 0.21 ^{bc}
	SNP		1.22 $\pm$ 0.12 ^{n-p}	2.67 $\pm$ 0.14 ^{9-h}	4.39 $\pm$ 0.17 ^{ef}	5.72 $\pm$ 0.19 ^b
	Control	29.90 $\pm$ 0.61 ^q	35.24 $\pm$ 0.67 ^{l-n}	40.36 $\pm$ 0.85 ^{cd}	44.16 $\pm$ 0.38 ^{bc}	46.59 $\pm$ 0.42 ^a
	XG 0.50 % + SNP		32.48 $\pm$ 0.52 ^{op}	34.67 $\pm$ 0.70 ^{m-o}	37.45 $\pm$ 0.48 ^{i-l}	39.51 $\pm$ 0.66 ^{f-i}
	XG 0.25 % + SNP		33.29 $\pm$ 0.90 ^{n-p}	36.18 $\pm$ 0.62 ^{j-r}	39.85 $\pm$ 0.63 ^{f-h}	42.44 $\pm$ 0.82 ^{c-d}
	XG 0.50 %		31.84 $\pm$ 0.81 ^{p-q}	35.62 $\pm$ 0.78 ^{k-n}	38.51 $\pm$ 0.90 ^{s-j}	41.62 $\pm$ 0.71 ^{d-f}
	XG 0.25 %		33.49 $\pm$ 0.75 ^{n-p}	37.65 $\pm$ 0.61 ^{h-k}	41.78 $\pm$ 0.93 ^{c-f}	43.91 $\pm$ 0.56 ^{b-d}
	SNP		34.17 $\pm$ 0.90 ^{m-p}	39.06 $\pm$ 0.56 ^{s-i}	42.34 $\pm$ 0.80 ^{c-e}	45.52 $\pm$ 0.74 ^{ab}
Respiration rate (mL CO₂ kg⁻¹ h⁻¹)	Control	0.0 $\pm$ 0 ⁱ	6.94 $\pm$ 1.13 ^{f-h}	13.88 $\pm$ 1.30 ^{cd}	19.44 $\pm$ 0.65 ^{ab}	23.61 $\pm$ 1.13 ^a
	XG 0.50 % + SNP		0.00 $\pm$ 0 ⁱ	4.16 $\pm$ 0.65 ^{s-i}	6.94 $\pm$ 1.31 ^{f-h}	9.72 $\pm$ 2.35 ^{d-f}
	XG 0.25 % + SNP		0.00 $\pm$ 0 ⁱ	8.33 $\pm$ 1.13 ^{e-g}	12.50 $\pm$ 2.35 ^{c-e}	15.27 $\pm$ 1.13 ^{bc}
	XG 0.50 %		0.00 $\pm$ 0 ⁱ	6.94 $\pm$ 1.73 ^{f-h}	9.72 $\pm$ 1.73 ^{d-f}	12.50 $\pm$ 2.36 ^{c-e}
	XG 0.25 %		2.77 $\pm$ 0.65 ^{hi}	9.72 $\pm$ 1.31 ^{d-f}	13.88 $\pm$ 1.73 ^{cd}	19.44 $\pm$ 0.65 ^{ab}
	SNP		5.55 $\pm$ 1.73 ^{f-h}	12.50 $\pm$ 0.65 ^{c-e}	15.27 $\pm$ 1.13 ^{bc}	20.83 $\pm$ 1.73 ^a
Spoilage (%)	Control	0.0 $\pm$ 0 ⁱ	6.94 $\pm$ 1.13 ^{f-h}	13.88 $\pm$ 1.30 ^{cd}	19.44 $\pm$ 0.65 ^{ab}	23.61 $\pm$ 1.13 ^a
	XG 0.50 % + SNP		0.00 $\pm$ 0 ⁱ	4.16 $\pm$ 0.65 ^{s-i}	6.94 $\pm$ 1.31 ^{f-h}	9.72 $\pm$ 2.35 ^{d-f}
	XG 0.25 % + SNP		0.00 $\pm$ 0 ⁱ	8.33 $\pm$ 1.13 ^{e-g}	12.50 $\pm$ 2.35 ^{c-e}	15.27 $\pm$ 1.13 ^{bc}
	XG 0.50 %		0.00 $\pm$ 0 ⁱ	6.94 $\pm$ 1.73 ^{f-h}	9.72 $\pm$ 1.73 ^{d-f}	12.50 $\pm$ 2.36 ^{c-e}
	XG 0.25 %		2.77 $\pm$ 0.65 ^{hi}	9.72 $\pm$ 1.31 ^{d-f}	13.88 $\pm$ 1.73 ^{cd}	19.44 $\pm$ 0.65 ^{ab}
	SNP		5.55 $\pm$ 1.73 ^{f-h}	12.50 $\pm$ 0.65 ^{c-e}	15.27 $\pm$ 1.13 ^{bc}	20.83 $\pm$ 1.73 ^a

coated fruits. Also on the last day of storage, the lower dose of coatings (XG 0.25 % + SNP) registered a higher value of respiration rate 42.44 ml CO₂ kg⁻¹ h⁻¹ in comparison with XG 0.50 % + SNP treatment (39.51 ml CO₂ kg⁻¹ h⁻¹). Also, at the end of the storage period, the maximum hike in respiration rate was observed in the control fruits which were 1.17-fold higher than the fruits coated with XG 0.50 % + SNP.

3.4. Spoilage (%)

The signs of spoilage by visual inspection were first detected in

control, SNP and XG 0.25 % treated fruits at the first interval while the other treated groups exhibited decay incidence on the second interval (Table 1). On the 8th day of storage, the spoilage incidence in XG 0.25 % + SNP coated fruits was 2.0 times higher than XG 0.50 % + SNP coated fruits. On the 12th day sole SNP treated fruits showed 12.50 % spoilage which was 3.00 and 1.80 –folds higher than XG 0.50 % + SNP and sole XG 0.50 % fruits. Also on the 16th day of storage period, spoilage in XG 0.50 % + SNP coated strawberries was 25.02 and 72.73% lower when compared to XG 0.50 % and SNP alone treatments. The maximum spoilage rate at the end of the experiment was attained by the control fruits (23.61 %) which was 2.42-folds higher than XG 0.50 % + SNP coated fruits.

3.5. SSC and TA (%)

The results recorded for SSC and TA of coated and control fruits are shown in Table 2. During initial sampling, the SSC level in all the

Table 2

Effect of different concentrations of XG and SNP (2.0 mM) on SSC, titratable acidity, TPC and TFC on strawberry fruits during storage. Data presented are the mean values $\pm$ SE of 3 replications ($p < 0.05$). Different letters indicate significant differences at ($p < 0.05$) using least significant difference.

Parameter	Treatment	Storage time (d)				
		0	4	8	12	16
SSC (°Brix)	Control	6.64 $\pm$ 0.05 ^p	7.12 $\pm$ 0.07 ^{j-n}	7.51 $\pm$ 0.06 ^{d-g}	8.08 $\pm$ 0.09 ^a	7.78 $\pm$ 0.07 ^{bc}
	XG 0.50 % + SNP		6.82 $\pm$ 0.09 ^{op}	6.93 $\pm$ 0.08 ^{m-o}	7.18 $\pm$ 0.07 ^{i-m}	7.25 $\pm$ 0.10 ^{s-k}
	XG 0.25 % + SNP		6.95 $\pm$ 0.04 ^{l-o}	7.08 $\pm$ 0.09 ^{j-o}	7.55 $\pm$ 0.06 ^{c-f}	7.45 $\pm$ 0.09 ^{d-g}
	XG 0.50 %		6.90 $\pm$ 0.08 ^{n-p}	7.21 $\pm$ 0.12 ^{b-l}	7.46 $\pm$ 0.09 ^{e-h}	7.40 $\pm$ 0.07 ^{e-i}
	XG 0.25 %		7.03 $\pm$ 0.10 ^{k-o}	7.33 $\pm$ 0.03 ^{f-j}	7.78 $\pm$ 0.08 ^{bc}	7.60 $\pm$ 0.07 ^{c-e}
	SNP		7.00 $\pm$ 0.06 ^{k-o}	7.40 $\pm$ 0.05 ^{e-i}	7.91 $\pm$ 0.07 ^{ab}	7.75 $\pm$ 0.08 ^{b-d}
	Control	1.54 $\pm$ 0.03 ^{e-h}	1.16 $\pm$ 0.03 ^{e-h}	0.97 $\pm$ 0.03 ^{j-n}	0.87 $\pm$ 0.02 ^{a-o}	0.82 $\pm$ 0.03 ^o
	XG 0.50 % + SNP	0.02 ^a	1.38 $\pm$ 0.03 ^{bc}	1.30 $\pm$ 0.02 ^{cd}	1.18 $\pm$ 0.03 ^{c-g}	1.11 $\pm$ 0.03 ^{f-i}
	XG 0.25 % + SNP		1.42 $\pm$ 0.03 ^b	1.22 $\pm$ 0.03 ^{d-f}	1.15 $\pm$ 0.04 ^{e-h}	1.05 $\pm$ 0.03 ^{h-k}
	XG 0.50 %		1.30 $\pm$ 0.02 ^{cd}	1.16 $\pm$ 0.02 ^{e-h}	1.07 $\pm$ 0.03 ^{s-j}	0.98 $\pm$ 0.04 ^{j-n}
	XG 0.25 %		1.23 $\pm$ 0.04 ^{de}	1.13 $\pm$ 0.03 ^{e-h}	0.99 $\pm$ 0.03 ^{j-m}	0.92 $\pm$ 0.03 ^{l-o}
	SNP		1.20 $\pm$ 0.02 ^{d-f}	1.01 $\pm$ 0.03 ^{i-l}	0.95 $\pm$ 0.03 ^{k-n}	0.89 $\pm$ 0.02 ^{m-o}
Titratable acidity (%)	Control	0.98 $\pm$ 0.01 ^a	0.75 $\pm$ 0.01 ^{ef}	0.60 $\pm$ 0.02 ^{hi}	0.44 $\pm$ 0.02 ^{kl}	0.36 $\pm$ 0.02 ^m
	XG 0.50 % + SNP		0.90 $\pm$ 0.01 ^b	0.77 $\pm$ 0.02 ^{d-f}	0.70 $\pm$ 0.03 ^{fg}	0.53 $\pm$ 0.03 ^{ij}
	XG 0.25 % + SNP		0.88 $\pm$ 0.02 ^{bc}	0.75 $\pm$ 0.02 ^{ef}	0.59 $\pm$ 0.01 ^{hi}	0.49 $\pm$ 0.01 ^{j-l}
	XG 0.50 %		0.87 $\pm$ 0.02 ^{bc}	0.71 $\pm$ 0.02 ^{fg}	0.62 $\pm$ 0.02 ^b	0.48 $\pm$ 0.01 ^{jk}
	XG 0.25 %		0.84 $\pm$ 0.01 ^{b-d}	0.66 $\pm$ 0.01 ^{gh}	0.53 $\pm$ 0.02 ^{ij}	0.44 $\pm$ 0.02 ^{kl}
	SNP		0.82 $\pm$ 0.02 ^{c-e}	0.62 $\pm$ 0.01 ^h	0.50 $\pm$ 0.01 ^{jk}	0.41 $\pm$ 0.02 ^{lm}
TPC (g kg⁻¹)	Control	1.32 $\pm$ 0.01 ^a	1.07 $\pm$ 0.01 ^{de}	0.96 $\pm$ 0.03 ^{s-i}	0.80 $\pm$ 0.01 ^{mn}	0.68 $\pm$ 0.02 ^o
	XG 0.50 % + SNP		1.24 $\pm$ 0.02 ^{ab}	1.17 $\pm$ 0.01 ^{bc}	1.05 $\pm$ 0.03 ^{ef}	0.93 $\pm$ 0.02 ^{h-j}
	XG 0.25 % + SNP		1.22 $\pm$ 0.01 ^{bc}	1.14 $\pm$ 0.02 ^{cd}	0.99 $\pm$ 0.04 ^{e-h}	0.89 $\pm$ 0.03 ^{i-l}
	XG 0.50 %		1.17 $\pm$ 0.02 ^{bc}	1.07 $\pm$ 0.02 ^{de}	0.91 $\pm$ 0.02 ^{h-k}	0.87 $\pm$ 0.02 ^{j-m}
	XG 0.25 %		1.18 $\pm$ 0.02 ^{bc}	1.02 $\pm$ 0.01 ^{e-g}	0.93 $\pm$ 0.02 ^{h-j}	0.81 $\pm$ 0.01 ^{l-m}
	SNP		1.14 $\pm$ 0.02 ^{cd}	0.98 $\pm$ 0.02 ^{fh}	0.84 $\pm$ 0.01 ^{k-m}	0.75 $\pm$ 0.04 ^{no}

treatments increased until the 12th day and thereafter declined while the TA of strawberry fruits showed a consistent decrease throughout the storage period. The SSC in the control sample increased from 6.64 to 8.08 % till the 12th day and then declined to 7.78 % at the end of the storage. On the other hand, XG 0.50 % + SNP coated fruits showed a continuous increase in SSC till the end of the storage period. On the 16th day, the fruit treated with XG 0.25% recorded higher SSC than the fruit coated with XG 0.25% + SNP. In comparison to sole XG 0.50 % or SNP treatments, combined XG 0.50 % + SNP treatment maintained significantly ($p < 0.05$) lower SSC levels of the strawberry fruit.

Throughout the storage period, the composite coatings maintained significantly higher TA than sole coatings and control. On the 8th day, XG 0.50 % + SNP treatment retained 11.38 and 25.10 % higher TA than sole XG 0.50 % and SNP treatment. Also, on the 8th day, alone SNP treated fruits recorded a value of 1.01 % which is 0.87 and 0.89 times higher than fruit treated with XG 0.50 and XG 0.25 %, respectively. On the 16th day, XG 0.50 % + SNP coated fruits exhibited 1.28 and 1.43 – fold higher acidity than SNP and control respectively.

3.6. Total phenols content (g kg^{-1})

The phenols content in both coated and uncoated fruits decreased during storage but the combined coated fruits maintained higher TPC than uncoated fruits (Table 2). The maximum accumulation of phenols on the 16th day was found in SNP enriched XG 0.50 % treatment followed by XG 0.25 % + SNP coated group with a value of 0.53 g kg^{-1} and 0.49 g kg^{-1} respectively. The rate of phenol decline in sole coated strawberries was more than SNP enriched XG coated fruits. On the 8th day in, XG 0.50 % + SNP treatment significantly ($p < 0.05$) retained 21.58 and 24.82 % higher TPC comparison to control and SNP, respectively. On the 12th day, the phenolic content in control strawberries reduced to 0.44 g kg^{-1} which was 12.76 and 45.61 % lower than SNP and XG 0.50 % + SNP treated fruits. Also on this day, XG 0.50 % + SNP treatment exhibited significantly higher ($p < 0.05$) TPC than sole XG 0.50 % and SNP treatment. At the end of the experiment, the highest and lowest TPC was recorded by XG 0.50 % + SNP and control strawberries having values of 0.53 g kg^{-1} and 0.36 g kg^{-1} respectively.

3.7. Total flavonoid content (g kg^{-1})

During storage, both treated and control fruits depicted a significant decrease in TFC (Table 2). A sharp decline of TFC in control fruits was found on 4th day which reached its lowest value of 0.68 g kg^{-1} at the end of the storage period. Whereas XG 0.50 % + SNP treatment retained the maximum amount of TFC and was 31.05 % higher than recorded in untreated strawberries. On 12th day, XG 0.50 % + SNP and XG 0.25 % + SNP samples maintained a higher level of TFC (1.05 and 0.99 g kg^{-1}) as compared to their sole coatings of XG 0.50 % and XG 0.25 %, respectively. Furthermore, SNP coated fruits registered 0.80 and 0.84-folds lesser TFC than XG 0.50 % + SNP and XG 0.25 % + SNP fruits.

3.8. Total anthocyanin content (mg kg^{-1})

As the storage period progressed, the TAC increased in all treatments till the 12th day and then started to decline (Table 3). Throughout the storage period, XG 0.50 % + SNP showed the least change in TAC as compared to other treatments. On the 8th day XG 0.25 % + SNP delayed the increase in TAC by showing a lower value of 654.3 mg/kg as compared to XG 0.50 % (670.7 mg kg^{-1}). At the last interval, the untreated fruits recorded the highest anthocyanin level was 727.9 mg kg^{-1} which was 17.59 % higher than XG 0.50 % + SNP treated fruits. Furthermore, XG 0.50 % + SNP coated fruits significantly maintained TAC than its sole XG and SNP coated fruits.

Table 3

Effect of different concentrations of XG and SNP (2.0 mM) on TAC, DPPH and FRAP activity on strawberry fruits during storage. Data presented are the mean values $\pm$ SE of 3 replications ($p < 0.05$). Different letters indicate significant differences at ($p < 0.05$) using least significant difference.

Parameter	Treatment	Storage time (d)				
		0	4	8	12	16
TAC (mg kg^{-1})	Control	515.4 $\pm$ 10.28 ^m	650.2 $\pm$ 9.18 ^{g-i}	729.8 $\pm$ 11.4 ^{bc}	786.1 $\pm$ 13.8 ^a	727.9 $\pm$ 12.1 ^{bc}
	XG 0.50 % + SNP		568.5 $\pm$ 13.2 ^l	637.1 $\pm$ 8.50 ^{h-j}	664.4 $\pm$ 15.6 ^{h-h}	610.2 $\pm$ 10.9 ^{jk}
	XG 0.25 % + SNP		590.8 $\pm$ 10.3 ^{kl}	654.3 $\pm$ 9.51 ^{g-i}	702.9 $\pm$ 9.25 ^{c-f}	638.3 $\pm$ 13.0 ^{h-j}
	XG 0.50 %		586.1 $\pm$ 11.6 ^{kl}	670.7 $\pm$ 12.2 ^{e-h}	727.5 $\pm$ 10.5 ^{bc}	655.1 $\pm$ 9.22 ^{g-i}
	XG 0.25 %		622.4 $\pm$ 9.30 ^{i-k}	683.9 $\pm$ 15.0 ^{d-g}	741.2 $\pm$ 11.6 ^{bc}	686.6 $\pm$ 13.2 ^{d-g}
	SNP		636.3 $\pm$ 12.5 ^{h-j}	711.2 $\pm$ 9.27 ^{cd}	762.2 $\pm$ 10.5 ^{ab}	708.4 $\pm$ 11.4 ^{c-e}
DPPH (% radical scavenging activity)	Control	83.34 $\pm$ 0.82 ^a	75.24 $\pm$ 1.20 ^{d-g}	68.92 $\pm$ 0.96 ^{ij}	64.88 $\pm$ 1.69 ^{kl}	60.95 $\pm$ 0.73 ^m
	XG 0.50 % + SNP		81.10 $\pm$ 1.05 ^{ab}	77.54 $\pm$ 1.11 ^{b-e}	74.31 $\pm$ 0.91 ^{e-g}	72.45 $\pm$ 1.38 ^{g-i}
	XG 0.25 % + SNP		78.94 $\pm$ 0.73 ^{b-d}	73.18 $\pm$ 1.22 ^{f-h}	72.01 $\pm$ 1.43 ^{g-i}	70.03 $\pm$ 0.98 ^{h-j}
	XG 0.50 %		80.09 $\pm$ 1.53 ^{a-c}	75.05 $\pm$ 0.93 ^{e-g}	69.96 $\pm$ 1.12 ^{h-j}	67.47 $\pm$ 1.00 ^{jk}
	XG 0.25 %		76.36 $\pm$ 0.46 ^{c-f}	71.99 $\pm$ 1.09 ^{g-i}	66.61 $\pm$ 1.32 ^{i-l}	64.34 $\pm$ 1.24 ^{k-m}
	SNP		77.36 $\pm$ 1.01 ^{b-e}	70.27 $\pm$ 0.89 ^{h-j}	67.83 $\pm$ 1.52 ^{jk}	62.97 $\pm$ 1.12 ^{lm}
FRAP (mmole kg^{-1})	Control	63.83 $\pm$ 0.91 ^a	56.36 $\pm$ 1.08 ^{d-f}	46.60 $\pm$ 1.58 ^{l-n}	40.85 $\pm$ 1.18 ^{o-q}	38.12 $\pm$ 1.33 ^q
	XG 0.50 % + SNP		61.88 $\pm$ 1.13 ^{ab}	58.91 $\pm$ 1.30 ^{b-d}	53.52 $\pm$ 1.07 ^{e-h}	49.17 $\pm$ 1.50 ^{i-l}
	XG 0.25 % + SNP		60.73 $\pm$ 1.28 ^{a-c}	55.76 $\pm$ 1.43 ^{d-g}	51.19 $\pm$ 1.14 ^{h-k}	47.58 $\pm$ 1.62 ^{k-n}
	XG 0.50 %		59.29 $\pm$ 1.05 ^{b-d}	52.81 $\pm$ 1.35 ^{f-i}	47.73 $\pm$ 1.85 ^{j-n}	44.21 $\pm$ 1.09 ^{n-p}
	XG 0.25 %		57.45 $\pm$ 1.30 ^{c-e}	51.94 $\pm$ 1.23 ^{g-j}	44.86 $\pm$ 1.47 ^{m-o}	41.87 $\pm$ 1.17 ^{o-q}
	SNP		56.96 $\pm$ 1.32 ^{c-f}	48.79 $\pm$ 1.09 ^{i-m}	43.94 $\pm$ 1.37 ^{n-p}	40.33 $\pm$ 1.18 ^{p-q}

3.9. Total antioxidant activity

The total antioxidant activity (TAA) in terms of DPPH and FRAP of strawberry fruits showed a declining trend throughout the storage period in all the treatments (Table 3). The highest DPPH and FRAP activity at the last interval were reported in XG 0.50 % + SNP samples (72.45 % RSA and $49.17 \text{ mmol kg}^{-1}$ respectively) followed by XG 0.25 % + SNP (70.03 % RSA and $47.58 \text{ mmol kg}^{-1}$ respectively) while the lowest TAA was recorded in control samples (60.95 % RSA and $38.12 \text{ mmol kg}^{-1}$ respectively). Furthermore, on the 8th day, the TAA of XG

0.25 % + SNP treated fruit was significantly higher in comparison with sole XG 0.25 % treated fruit with TAA of 73.18 % RSA and 55.76 mmol kg⁻¹ and 71.99% RSA and 51.94 mmol kg⁻¹ respectively. On the 12th day, the TAA of SNP combined XG 0.50 % samples (74.31 % RSA and 53.52 mmol kg⁻¹ respectively) was significantly ($p < 0.05$) higher as compared to control samples (64.88 % RSA and 40.85 mmol kg⁻¹ respectively). At the end of storage, TAA in pure SNP samples in terms of DPPH and FRAP was 0.86 and 0.82 folds lower than XG 0.50 % + SNP samples.

3.10. Antioxidant defense enzyme activities

The trend in antioxidant defense enzyme activities as a function of different treatments and storage time was declining but the coated samples retained higher levels of enzymatic activities as compared to the uncoated samples (Table 4).

A consistent increase in PAL activity was observed in all the coated and uncoated strawberries, followed by a decline. PAL activity in XG 0.50 % + SNP treated fruits peaked on the 12th day having a value of 4.17 U mg⁻¹ protein whereas untreated strawberries reached its peak on the 8th day, recording a value of 3.30 U mg⁻¹ protein. On the 12th day, XG 0.50 % + SNP coated fruits recorded 9.02 % higher PAL activity as compared to sole XG 0.50 % coated fruits. At the last interval, XG 0.50 % + SNP held the highest activity of 3.78 U mg⁻¹ protein which was significantly ($p < 0.05$) higher by 37.73 % and 50.74 % higher when compared to sole SNP (2.58 U mg⁻¹ protein) and control (2.25 U mg⁻¹ protein), respectively. CAT activity followed a declining trend throughout the storage period irrespective of the treatments. However, coated fruits exhibited higher CAT activity in comparison to uncoated fruits. On the 8th day of storage, the untreated strawberries exhibited a 24.45 U mg⁻¹ protein activity and it was 0.86 and 0.94 times lower than XG 0.50 % + SNP and XG 0.50 % coated fruits. On the 12th day, SNP samples showed 1.03 fold higher CAT activity of 23.24 U mg⁻¹ protein in comparison with the control. At the end of the storage period, XG 0.25 % + SNP treated fruits recorded a significantly higher value of CAT activity (23.59 U mg⁻¹ protein) while sole XG 0.25 % treated fruits recorded 22.86 U mg⁻¹ protein activity. Also, the highest and the lowest CAT activity was observed in XG 0.50 % + SNP and control samples attaining a value of 24.58 and 21.29 U mg⁻¹ protein respectively.

POD activity exhibited an initial increase followed by a gradual decline. Control, SNP and XG 0.50 % treatments marked their peak on the 8th day with a value of 2.68, 2.86 and 3.24 U mg⁻¹ protein respectively and declined thereafter. On the 12th day, the maximum POD activity (3.42 U mg⁻¹ protein) was recorded by XG 0.50 % + SNP treated strawberries followed by XG 0.25 % + SNP treatment (3.24 U mg⁻¹ protein) which afterwards entered the declining phase. At the end of the storage, the lowest POD activity was attained in the uncoated strawberries (2.19 U mg⁻¹ protein) followed by SNP treated fruits (2.38 U mg⁻¹ protein) which was 1.38 and 1.27- folds higher than XG 0.50 % + SNP treatment (3.03 U mg⁻¹ protein). Furthermore, the sole coated XG fruits exhibited lower enzymatic activity as compared to their counter combined coatings. Thereby, XG + SNP coated fruits exhibited significantly ($p < 0.05$) higher antioxidant enzymatic activity than their sole XG or SNP counterparts.

3.11. PPO activity

As the duration of storage was prolonged, PPO activity showed a consistent increase in coated and uncoated strawberries (Table 4). The maximum change in PPO activity from 1.25 U mg⁻¹ protein (day 0) to 2.14 U mg⁻¹ protein (day 4) was found in control samples. Throughout the storage period, SNP enriched XG coatings delayed the increase in PPO activity in comparison to their sole coatings. On the 12th day of storage, XG 0.50 % + SNP and XG 0.25 % + SNP samples recorded a PPO activity of 2.35 and 2.67 U mg⁻¹ protein which was 19.57 % and 10.97 % lower than sole XG coated samples. Also, XG 0.50 % exhibited 4.10 %

Table 4

Antioxidant (POD, CAT, PAL) and PPO enzyme activity of strawberry fruits uncoated or coated with different concentration of XG and SNP (2.0 mM) during storage. Data presented are the mean values $\pm$ SE of 3 replications ($p < 0.05$). Different letters indicate significant differences at ($p < 0.05$) using least significant difference.

Parameter	Treatment	Storage time (d)				
		0	4	8	12	16
POD (U mg ⁻¹ protein)	Control	2.24 $\pm$	2.57 $\pm$	2.68 $\pm$	2.45 $\pm$	2.19 $\pm$
			$\pm$	0.11 ^{e-h}	0.09 ^{e-h}	0.09 ^j
	XG 0.50 %	0.07 ^{ij}	2.80 $\pm$	3.26 $\pm$	3.42 $\pm$	3.03 $\pm$
	+ SNP		0.05 ^{d-f}	0.10 ^{ab}	0.08 ^a	0.07 ^{b-d}
	XG 0.25 %		2.73 $\pm$	3.13 $\pm$	3.24 $\pm$	2.86 $\pm$
	+ SNP		0.08 ^{d-g}	0.10 ^{a-c}	0.09 ^{ab}	0.11 ^{c-e}
	XG 0.50 %		2.69 $\pm$	3.24 $\pm$	3.04 $\pm$	2.68 $\pm$
			0.06 ^{e-h}	0.12 ^{ab}	0.08 ^{b-d}	0.08 ^{e-h}
	XG 0.25 %		2.61 $\pm$	3.02 $\pm$	2.85 $\pm$	2.62 $\pm$
			0.07 ^{e-h}	0.06 ^{b-d}	0.13 ^{c-f}	0.08 ^{e-h}
CAT (U mg ⁻¹ protein)	Control	30.51 $\pm$	27.12 $\pm$	24.45 $\pm$	22.38 $\pm$	21.29 $\pm$
		0.36 ^a	0.60 ^{d-g}	0.49 ^{j-m}	0.27 ^{op}	0.36 ^p
	XG 0.50 %		29.36 $\pm$	28.16 $\pm$	26.72 $\pm$	24.58 $\pm$
	+ SNP		$\pm$	$\pm$	$\pm$	$\pm$ 0.40 ^{i-l}
	XG 0.25 %		0.49 ^{ab}	0.58 ^{b-e}	0.50 ^{e-h}	23.59 $\pm$
	+ SNP		28.53 $\pm$	27.28 $\pm$	26.51 $\pm$	$\pm$ 0.29 ^{l-o}
			$\pm$	$\pm$	$\pm$	$\pm$ 0.29 ^{l-o}
	XG 0.50 %		1.10 ^{b-d}	0.30 ^{d-f}	0.30 ^{f-i}	23.08 $\pm$
			28.84 $\pm$	25.78 $\pm$	25.13 $\pm$	23.08 $\pm$
			$\pm$	$\pm$	$\pm$	$\pm$ 0.09 ^{m-o}
PAL (U mg ⁻¹ protein)	Control	1.90 $\pm$	2.52 $\pm$	3.30 $\pm$	2.96 $\pm$	2.25 $\pm$
		$\pm$	0.12 ^{ij}	0.06 ^{d-g}	0.21 ^{g-i}	0.11 ^k
	XG 0.50 %	0.11 ^k	3.30 $\pm$	3.95 $\pm$	4.17 $\pm$	3.78 $\pm$
	+ SNP		0.10 ^{d-g}	0.06 ^{ab}	0.19 ^a	0.13 ^{a-d}
	XG 0.25 %		3.12 $\pm$	3.72 $\pm$	3.9 $\pm$	3.49 $\pm$
	+ SNP		0.07 ^{e-h}	0.22 ^{a-e}	0.12 ^{a-c}	0.20 ^{b-f}
	XG 0.50 %		2.87 $\pm$	3.53 $\pm$	3.81 $\pm$	3.26 $\pm$
			0.15 ^{g-i}	0.06 ^{b-f}	0.15 ^{a-c}	0.08 ^{e-g}
	XG 0.25 %		2.94 $\pm$	3.84 $\pm$	3.51 $\pm$	2.86 $\pm$
			0.08 ^{hi}	0.20 ^{a-c}	0.12 ^{b-f}	0.07 ^{g-i}
PPO (U mg ⁻¹ protein)	Control	1.25 $\pm$	2.14 $\pm$	2.94 $\pm$	3.87 $\pm$	4.17 $\pm$
		$\pm$	0.10 ^{j-m}	0.12 ^{f-h}	0.13 ^{ab}	0.14 ^a
	XG 0.50 %	0.10 ^o	1.71 $\pm$	1.85 $\pm$	2.35 $\pm$	3.18 $\pm$
	+ SNP		0.10 ^{m-o}	0.17 ⁱ⁻ⁿ	0.13 ^{j-k}	0.15 ^{d-g}
	XG 0.25 %		1.57 $\pm$	2.05 $\pm$	2.67 $\pm$	3.37 $\pm$
	+ SNP		0.15 ^{no}	0.14 ^{k-m}	0.12 ^{hi}	0.11 ^{c-f}
	XG 0.50 %		1.86 $\pm$	2.28 $\pm$	2.86 $\pm$	3.42 $\pm$
			0.11 ^{l-n}	0.09 ^{i-l}	0.17 ^{gh}	0.13 ^{b-e}
	XG 0.25 %		1.92 $\pm$	2.54 $\pm$	2.98 $\pm$	3.63 $\pm$
			0.13 ^{k-n}	0.17 ^{h-j}	0.17 ^{e-h}	0.13 ^{b-d}
SNP			2.02 $\pm$	2.71 $\pm$	3.51 $\pm$	3.83 $\pm$
			0.11 ^{k-n}	0.13 ^{hi}	0.20 ^{b-d}	0.12 ^{a-c}

lower PPO activity in comparison to XG 0.25 % coatings. On the 16th day, pure SNP treated strawberries recorded PPO activity of 3.83 U mg⁻¹ protein which was 18.54 and 12.77 % higher than XG 0.50 % + SNP and XG 0.25 % + SNP samples. The highest and lowest browning enzymatic activity on the last day of the experiment was recorded in control and XG 0.50 % + SNP samples (4.17 and 3.18 U mg⁻¹ protein respectively) differencing by 26.93 %.

3.12. Correlation analysis

The extent of correlation among various antioxidant parameters

during storage was studied using Pearson's correlation analysis (Fig. 3). Spoilage in strawberries represented a negative correlation with TA ($r = -0.88$), TPC ($r = -0.91$), TFC ($r = -0.91$), DPPH ($r = -0.90$), FRAP ($r = -0.90$) and CAT ($r = -0.89$) activity, indicating that spoilage increased with decrease in the activity of antioxidant compounds and the latter has a role in fruit defense mechanism. Moreover, TA showed a positive correlation with all the antioxidant compounds, indicating their presence in maintaining the quality of strawberries. TPC maintained a highly positive correlation with DPPH ($r = 0.92$), FRAP ($r = 0.93$) and CAT ($r = 0.94$), while a less negative correlation with PAL ($r = -0.45$) and POD ($r = -0.24$), indicating its contribution in antioxidant activity and that synthesis of TPC via phenylpropanoid pathway continued to occur even after the fruit harvest for a short period. Similarly, TFC showed less negative correlation with PAL ($r = -0.40$). The antioxidant capacity detected by DPPH and FRAP was strongly correlated with TPC, TFC and CAT, while it was less negatively correlated with PAL and POD. These results suggest that TPC, TFC, and CAT are the major contributors to the antioxidant activity in stored strawberries while PAL and POD also played a significant role in maintaining the quality and antioxidant defense system.

4. Discussion

This study investigated the effects of XG with SNP treatment on postharvest decay and storage quality of Winter Dawn strawberry fruit. The possible chemical and structural interactions of the different formulations of edible coatings applied on strawberry fruit were studied using FT-IR spectroscopy. The FT-IR spectrum of all the coating displayed a broad transmission band at 3268 cm^{-1} attributed to the axial deformation of the OH group. Slight vibrations in the region $920\text{--}1100\text{ cm}^{-1}$ could be detected which presents a typical profile of polysaccharides (CC/CO bond stretching) (Shalviri, Liu, Abdekhodae, &

Wu, 2010). In all the xanthan gum coatings, vibrations in the range $1600\text{--}1700\text{ cm}^{-1}$ are due to the O-H angular deformation and symmetric stretching of the carboxylate of pyruvates and glucuronic acid along with the CO stretching of alkyl esters (Bajaj, Kumar, Gill, Jawandha, & Kaur, 2024). Furthermore, bands around 1791 cm^{-1} indicate axial deformation of C=O ester, carboxylic acid, aldehydes and ketones; band at 1636 cm^{-1} can be assigned to the axial deformation of C=O from enols (-diketones) (Hazirah, Isa, & Sarbon, 2016). Also, a weak band at 1418 cm^{-1} corresponds to the asymmetric stretching vibration of carboxylate ion while the vibrations in the range $1340\text{--}1373\text{ cm}^{-1}$ is attributed to the bending mode of $-\text{CH}_2$ (Thombare, Mahto, Singh, Chowdhury, & Ansari, 2023). In SNP coating clear characteristic peak of N=O at 1943 cm^{-1} could be observed, which confirms the release of NO by SNP (Luo et al., 2022). Also, in composite coating decrease in spectral band at 1941 cm^{-1} could be detected which indicates the successful incorporation of SNP into xanthan gum.

Our research results showed that the XG + SNP treatment can retain strawberry fruit quality by eliciting the antioxidant defense system. Weight loss in fresh fruit is an important parameter in evaluating post-harvest quality. The present results depict a minimum weight loss in XG 0.50 % + SNP 2.0 mM as compared to other treatments. The composite coating provides a protective barrier against water loss from the fruit surface, whereas a dry atmosphere around uncoated fruits results in maximum water loss (Prasad et al., 2022). SNP is the donor of NO on the surface of the fruit and has the potential to deter respiration rate, thus delaying the ripening process (Adhikary et al., 2021). The results of our study are in line with the findings of Kumar, Upadhyay, Trajkovska-Petkoska, Gniewosz, and Kieliszek (2023), who reported reduced weight loss in XG-based composite-coated mango fruit. Furthermore, Kaur et al. (2022) also reported that the application of edible coatings functionalized with SNP reduced the weight loss in lemon fruits during their postharvest storage. The respiration rate of fruit indicates their metabolic rate, which is a crucial physiological parameter. Increased respiration rates lead to early senescence as well as the deterioration of fruit quality. A delay in the rate of CO_2 evolution was recorded in the strawberries coated with XG incorporated with SNP during storage. The exogenous application of NO extended fruit storage life by deterring respiration rates, which delayed the ripening process and prolonged fruit storage life (Saba & Moradi, 2017). The synergistic effect of XG and SNP has the ability to form a semipermeable barrier for the exchange of gas or moisture between the surrounding environment and the fruit, which reduces O_2 permeability and the emission of CO_2 from the fruit. Our results agree with the previous study of Kumar, Upadhyay, et al. (2023), who reported that the postharvest application of XG functionalized with pomegranate peel extract was effective in reducing the respiration rate in coated mango fruit. Additionally, we observed that the inhibitory effect of XG 0.50 % + SNP on the weight loss and respiration rate was more pronounced than sole XG or SNP treatments, suggesting a synergistic effect. In our study, composite coatings of XG 0.50 % + SNP significantly ($p < 0.05$) prevented the rise in SSC content as compared to all the treatments at the end of storage. Thus, XG 0.50 % + SNP treatment showed a synergistic effect. The loss of moisture during storage is one of the factors causing the fruit's SSC to rise (Ali et al., 2021). Furthermore, complex organic metabolites are converted into simple molecules and other metabolic activities can occur, including the hydrolysis of starch into sugars, which leads to an increase in SSC in fruits during storage (Xu et al., 2022). The application of coating materials like XG on the fruit surface minimizes the breakdown of complex sugars or starch and delays the increase in TSS concentration of 'Thompson Seedless' table grapes over a storage period (Baraiya, Ramana Rao, & Thakkar, 2016). A decreasing trend in the TA was observed in both the treated and untreated fruits attributing to the conversion of organic acids into other primary substrates. However, the composite coating of XG + SNP worked synergistically to retain the highest TA content over the storage period compared to sole XG, SNP and control treatments. Arguably, composite coatings created a superior

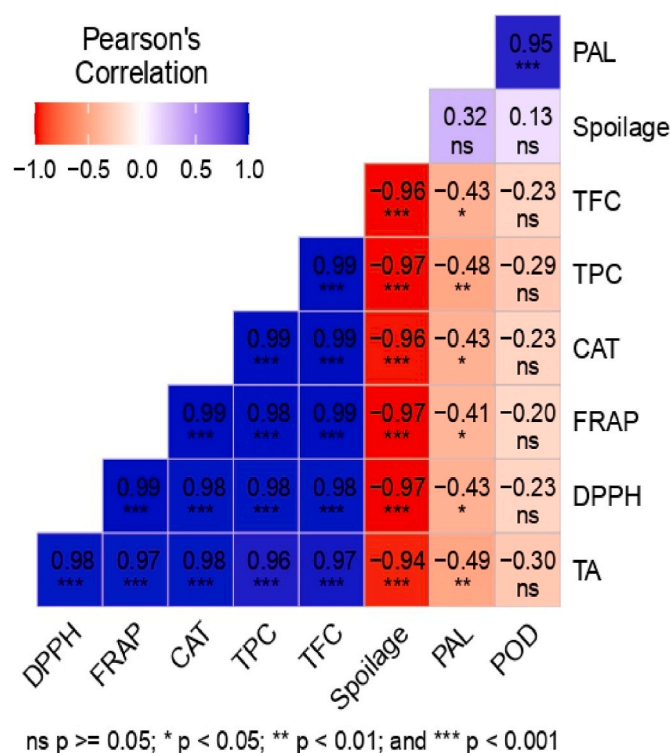


Fig. 3. Pearson's correlation analysis between the various antioxidant related parameters in strawberry fruits uncoated and coated with different concentrations of xanthan gum and sodium nitroprusside. Blue colors indicate a positive correlation while red colors indicate a negative correlation.

semi-permeable barrier around the fruit that could have reduced the gaseous exchange and delayed the oxidative degradation of organic acids present in the fruit (Golly et al., 2019). Our results are in accordance with the findings of Sharma and Rao (2015) in fresh-cut pear, where composite xanthan gum plus cinnamic acid coating recorded minimum loss in TA and SSC content. Similar results were also reported by Sahu et al. (2020) in guava fruits treated with SNP retained higher TA and SSC content as compared to control fruit during storage.

Plants are considered as biofactories for the production of secondary metabolites including phenolic compounds that may not be necessary for basic life processes, but has a crucial role in alleviating various abiotic and biotic stresses. Phenol and flavonoids are capable of directly quenching the ROS by donating the hydrogen atom or inhibiting free radicals by chelating free metal ions (Cheynier, Comte, Davies, Lattanzio, & Martens, 2013; Lu et al., 2021). Furthermore, secondary metabolites are involved in pathogen-related defense mechanism and ensure the nutritional quality of postharvest fruits during storage (Ze, Gao, & Jiang, 2022). In our study, regardless of the coating treatment, the total flavonoid, phenol, anthocyanin contents and antioxidant activity content of the fruits decreased during the extension of the storage period. However, XG + SNP treatment retained the higher TPC and TFC as compared to sole XG, SNP and control, thereby showing a synergistic effect with each other. Furthermore, the degradation of TAC was highly prevented by XG + SNP treatments than sole XG, SNP and control treatments. This confirms the synergistic behaviour of XG and SNP coatings. The anthocyanin synthesis occurs during chlorophyll degradation, which indicates a close relationship between chlorophyll and anthocyanin metabolism (Li et al., 2019). Moreover, anthocyanin formation continues after harvesting through the flavonoid and anthocyanin pathways (Nguyen, Nguyen, & Nguyen, 2020). In fruit, anthocyanins function as phytoalexins, antioxidants, and key components of plant defense mechanisms against microbial infections (Li et al., 2019). Various phenolic compounds in fruits, such as flavonoids contribute to their antioxidant properties (Grobela, Kalisz, & Kieliszek, 2019). These compounds directly quench ROS and serve as strong non-enzymatic antioxidant metabolites protecting cells against oxidative damage (Golly et al., 2019). Polyphenolic compounds are predominantly responsible for antioxidant activity (Kalisz et al., 2020) DPPH and FRAP are indexes that measure tissue antioxidant capacity and our results also depict a strong correlation of TPC and TFC with antioxidant activity. However, these phenolic compounds are subjected to oxidation reactions by air or PPO enzyme. The synergistic effect of XG and SNP coatings might be capable of preventing gaseous exchange during storage, reducing the amount of O₂ available for the fruit tissues to breathe, significantly delaying the respiratory rate in coated fruit and retaining higher TPC, TFC and antioxidant activity as compared to uncoated fruits. On the surface of the fruit, SNP is a source of NO and highly reactive species that act as antioxidants, so it may also contribute to the preservation and retention of high levels of antioxidant activity (Adhikary et al., 2021). The combined coating of XG loaded with *Lactobacillus plantarum* preserved the antioxidants and bioactive compounds in fresh-cut cantaloupe and honeydew melons, thereby reducing the postharvest losses (Chikhala, Seke, Slabbert, Sultanbawa, & Sivakumar, 2024). Furthermore, the ability of XG to form a protective barrier on fruit surfaces reduces the permeability of the surface (Kumar, Upadhyay, TrajkovskaPetkoska, Gniewosz, & Kieliszek, 2023). These results are in line with the outcomes of (Golly et al., 2019) who reported that table grapes coated with an XG-based coating retained a higher non-enzymatic antioxidant defense system during their postharvest storage. Our finding suggests that the composite coatings made from XG and SNP reduced the spoilage in coated fruits. SNP in postharvest treatment has been used as a donor agent of NO which contributed to the enhanced resistance against disease and the suppression of respiration and ethylene production during storage (Zhu & Zhou, 2007). The SNP treatments maintain fruit quality by suppressing ROS production and preventing pathogenic fungal decay by modulating antioxidant and

antifungal metabolic pathways (Sahu et al., 2020; Siddiqui et al., 2011). The results of these studies confirmed previous findings that SNP effectively retards decay loss through increased enzymatic and non-enzymatic defense mechanisms in response to NO and maintains quality traits in guava (Sahu et al., 2020) and suppresses anthracnose decay in mango (Ren et al., 2020). In our study, XG + SNP coated fruits maintained the higher antioxidant enzymatic activities viz. PAL, CAT and POD at the end of the storage period as compared to sole XG, sole SNP and control fruits. Therefore, XG + SNP coatings showed a synergistic effect. The PAL enzymes promoted the production of phenolic and flavonoids content via the phenylpropanoid pathway, and phenols helped enhance the secondary plant metabolic pathway of defense (Ge et al., 2019). Further, XG coating can act as a superior barrier against the permeability of water and gases, which helps to maintain PAL activity in XG-coated fruit and may contribute to a higher TPC than uncoated fruit (Galindo-Pérez, Quintanar-Guerrero, Mercado-Silva, Real-Sandoval, & Zambrano-Zaragoza, 2015). SNP continuously release NO on the surface of the fruit, which might contribute to increased accumulation of phenols and flavonoids via upregulation of defense enzymes such as PAL and POD in response to the antioxidant properties of NO. This could also be another mechanism for maintaining higher antioxidant enzyme activity (Ge et al., 2019). In our study, the combination of XG and SNP delayed the rise in PPO activity during storage as compared to sole XG and SNP treatments, whereas PPO activity in control fruits increased at a fastest pace. Therefore, XG + SNP combined treatment showed a synergistic effect. PPO enzyme activity is chiefly responsible for enzymatic browning symptoms in fruits. Enzymatic browning is mainly caused by the interaction between phenolic substrates and PPO enzyme. Disintegration and decompartmentalization of cellular membranes during ripening lead to the enzymatic oxidation of o-diphenols to o-quinones. The browning of fruit is caused by the further polymerization of o-quinones into polyphenols (Sinha et al., 2021). The results from the present study confirm the findings of (Ge et al., 2019), which showed that blueberry fruit immersed in SNP solution maintained PPO activity and conserved fruit quality during the storage period. Fruit treated with composite XG and SNP coatings increased the accumulation of antioxidant activity, which improved POD enzyme activity compared to uncoated fruit in the late stage of storage. During the defense reaction against pathogenic agents, the POD enzyme participates in wall-building processes by oxidizing phenols, suberization, and lignification of host cell walls. These observations were consistent with earlier studies that found XG-enriched cinnamic acid coating delayed fruit senescence by retaining maximum POD activity during the storage period in fresh-cut pears (Sharma & Rao, 2015). CAT is a key enzyme that catalyzes the conversion of hydrogen peroxide molecules into water and oxygen, enabling plants to carry out their cellular processes (Meitha, Pramesti, & Suhandono, 2020). Our results revealed that XG 0.50 % + SNP 2.0 mM coated fruit registered higher activities of CAT than uncoated fruit at the end of storage. These results confirm the findings of Ge et al., 2019, who reported higher CAT activity in SNP immersed blueberries. Also, jujube fruits coated with XG delays ripening and showed higher activities of CAT and POD enzymes along with reducing the activity of fruit softening enzymes (Naveed, Nawaz, Ali, & Ejaz, 2024). Hence, by combining the gas barrier effect of XG along with the antioxidant signaling potential of SNP, the quality of strawberry fruits stored at low temperature can be maintained for 16 days.

5. Conclusion

In this study, the combined treatment of xanthan gum 0.50 % and sodium nitroprusside 2.0 mM was the most effective treatment among all the tested coatings for maintaining the quality of strawberry fruit for 16 days under low temperature storage. The XG 0.50 % + SNP 2.0 mM coated fruits showed a reduction in weight loss, respiration rate and spoilage as compared to other treatments. The functionalized coating also sustained significantly higher TA in fruits with lower SSC and TAC

values. The edible coating accumulated a higher content of secondary metabolites viz. TPC and TFC which contributes to increase fruit quality and help in combating various abiotic and biotic stresses. Moreover, the coated fruit also exhibited retarded PPO enzyme activity and retained a higher activity of enzymes, viz., CAT, POD, and PAL. Due to the biodegradable nature and easy application, of this edible coating, it can be used at commercial scale to maintain the postharvest quality and to regulate the fresh strawberry market. Overall, the results suggested that the application of novel composite coating of XG 0.50 % + SNP 2.0 mM can be used to maintain the quality and postharvest life of strawberries by eliciting its role in creating gas barrier effect and the activation of antioxidant metabolism. Future research needs to investigate the further characterization of coatings and in-depth the microbial decay of fruit coated with XG coatings functionalized with SNP whilst in storage.

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CRediT authorship contribution statement

Anuja Gautam: Writing – original draft, Investigation, Formal analysis, Data curation. **P.P.S. Gill:** Writing – review & editing, Conceptualization. **NavPrem Singh:** Resources. **S.K. Jawandha:** Supervision, Resources. **Rachna Arora:** Resources. **Arashdeep Singh:** Resources. **Ajay:** Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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